

Epilepsy Detection From EEG Using Complex Network Techniques: A Review

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(Methodological Review)

Abstract—Epilepsy is one of the most chronic brain disorder recorded from since 2000 BC. Almost one-third of epileptic patients experience seizures attack even with medicated treatment. The menace of SUDEP (Sudden unexpected death in epilepsy) in an adult epileptic patient is approximately 8–17% more and 34% in a children epileptic patient. The expert neurologist manually analyses the Electroencephalogram (EEG) signals for epilepsy diagnosis. The non-stationary and complex nature of EEG signals this task more error-prone, time-consuming and even expensive. Hence, it is essential to develop automatic epilepsy detection techniques to ensure an appropriate identification and treatment of this disease. Nowadays, graph-theory has been considered as a prominent approach in the neuroscience field. The network-based approach characterizes a hidden sight of brain activity and brain-behavior mapping. The graph-theory not even helps to understand the underlying dynamics of EEG signals at microscopic, mesoscopic, and macroscopic level but also provide the correlation among them. This paper provides a review report about graph-theory based automated epilepsy detection methods. Furthermore, it will assist the expert's neurologist and researchers with the information of complex network-based epilepsy detection and aid the technician for developing an intelligent system that improving the diagnosis of epilepsy disorder.

Index Terms—Complex network, Classification, EEG, EEG signal, feature extraction, Graph Theory, Machine learning.

I. INTRODUCTION

EPILEPSY has been recognized as one of the most chronic brain dysfunction among neurological diseases [1], [2]. Around 50 million population at the world level are suffered from this epilepsy syndrome [3]. According to the World health organization, the epilepsy occurrence rate per year is 2.4 million, and it can affect any age group. The premature death rate of epileptic patients is two to three times higher than the normal

healthy person [4]. When there is a manifestation of more than two seizure attack concurrently, then epilepsy is declared [5]. During epileptic seizure attack, there is the incidence of abnormal electrical action in the brain because of the disparity of excitatory and inhibitory synapses present in the brain [6], [7]. Computerized tomography scan, Magnetic resonance imaging, Functional MRI and High-density Electroencephalogram (EEG) are the names of some tools which helps in the diagnosis of epilepsy. EEG has been recognized as one of the best medical test which assists in epilepsy diagnosis because EEG data exhibit high temporal and poorly spatial resolution [8], [9]. Also, the epileptiform seizure activity can be clearly observed in the EEG of epileptic patient even in the dearth of an epileptic-seizure attack [10]. Generally, the clinicians visually examine the EEG recording to detect epilepsy [11]–[13]. The visual EEG inspection is tiresome, fallible and even not cost effective because the cost of studying the EEG is high [14]–[17], [105]–[106]. Finding traces of seizure activity by experts' neurologist through visual examination of EEG is a challenging issue and cannot be considered as a very reliable procedure [18], [19]. In addition to this, EEG analysis not only assisting in the diagnosis of anti-epileptic medication but also plays a crucial role in all the neuro-surgery related to epilepsy. As a result, there is continuously an obligatory of automated seizure detection techniques that assist the clinicians for the diagnosis of epilepsy and also reduce cost and time.

Research in the arena of automatic epilepsy detection techniques started in 1982 when Gotman [20] first time proposed an automatic method to detect the epileptic seizure and performed some experiments wherein the amplitude of EEG signals was measured with respect to the background, the period of time and the periodicity of EEG signals. However, the proposed algorithm was unsuccessful to identify seizure activity from that EEG signals wherein the frequency bustle is high, and amplitude is low. Later on, different techniques were proposed by different researchers for automated detection of epilepsy from EEG signals. Fourier Transform is one of the popular techniques to detect epilepsy [21]–[25] but suffered from great noise sensitivity and lack of spectral estimation. Wavelet Transform based seizure detection techniques [26]–[30] have the drawback of selection of an appropriate mother wavelet. Time-frequency distribution based methods [31]–[37] has limitations of slow in speed as the time for computing the gradient ascent is high and extracted measures are inter-dependent. Some researchers

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had proposed seizure epileptic seizure detection methods based upon the parametric analysis. Like Granger Causality (GC) parameter was used to identify epileptic seizure from EEG [38]–[41]. Even though GC is easy to implement but it has the drawback of sensitivity towards volume conduction and noise. Phase-based techniques [42]–[45] has the limitation of uncertain results with the wrong choice of the phase difference. The traditional approaches have some restrictions and drawbacks like the non-parametric approach has the limitation of spectral leakage where the level of the EEG signal reduces, and the consequences can be easily shown in the whole frequency spectrum. The parametric approach has the drawback of the absence of time mechanism of EEG signals whereas, the time component provides beneficial information about EEG signals. Power spectrum and autocorrelation functions are not able to give any phase information regarding EEG signals. The major drawback of applying the recurrence plot, RPDE and RQA is that the choice of recurrence threshold value (ϵ) to generate recurrence plot which covers sufficient recurrence points so that the generated recurrence plot provides enough and valuable information about the EEG signals [101]. Due to the limitations of the traditional approaches for the analysis and processing of EEG data to detect epileptic seizure, the graph theory has become one of the key research field in epilepsy detection.

Even though early discoveries in the neuroscience research also suggest the importance of graph theory in clinical interpretability [46]–[50], [96]. Graph theory harvests important information about the underlying brain connectome with the help of certain topological properties of the EEG signals network [51], [52]. The statistical features of the network build from EEG signals provide critical knowledge about dysfunction related to the structure and function of the brain with epilepsy. In addition to that, research study covered from [65]–[100], highlight the significance of the graph theory or complex network in epilepsy detection.

From the last few years, complex network-based EEG signals analysis for the detection or prediction of epileptic seizure are showing vast progress. By considering this information, in this research study, we review the emergent research work on graph theory-based techniques to detect epilepsy from EEG signals. Section II describes several graphical parameters that are used as feature vector in the analysis and classification of epileptic EEG signals. In Section III, we present the overview about the most important approaches for converting EEG signals to complex network, i.e., Visibility Graph Approach, Horizontal Visibility Graph Approach, Time Series' Complex Network (TSCN) Approach, Graph-Theory based approaches, and Weighted Graph-based approaches. The research work related to epileptic seizure detection based upon these above five approaches are also included in this section. Section IV includes discussion and perspectives. Finally, the concluding remarks are drawn in Section V.

II. PROPERTIES OF NETWORK

This section describes about the different measures of the complex network or graph that are used (by the techniques

discussed in section III) as an extracted feature for the epilepsy detection. These graphical parameters are mined as a feature extraction part and used for the classification of different types of EEG signals. Below is the brief information and definition about that:

- a) *Degree Distribution (DD)*: The degree of a node (vertex) of the graph $G(V, E)$ is defined as the total number of links (associates) that particular vertex have with the other vertices in the graph or network.

The Degree distribution of a network or graph $P_{degree}(n)$ is a probability distribution function which is defined as the fraction of vertex in the network with degree n [65]. The below (1), elucidate the probability of vertex degrees in the graph $G(V, E)$.

$$P_G(n) = P(D_{degree}(v) = n); v \in V(G), \quad (1)$$

In the above equation, $D_{degree}(v)$ represents the degree of the vertex v and $P_G(n)$ denotes the probability with the degree of the vertex is equivalent to n .

- b) *Graph Index Complexity (GIC)*: The GIC measure is to analyze the graph. It provides the information about the link amongst the vertices of the graph. GIC is expressed as [66]:

$$C_k = 4c(1 - c), \quad (2)$$

where,

$$c = \frac{k - 2 \cos\left(\frac{\pi}{N+1}\right)}{N - 1 - 2 \cos\left(\frac{\pi}{N+1}\right)}, \quad (3)$$

In the above equations, k symbolizes the highest value eigenvalue correspond to the adjacency matrix of a network having N number of vertices.

- c) *Average Clustering Coefficient (A_{VGC})*: The Clustering Coefficient is a statistical feature of a graph that quantify the degree to which the node have a tendency to group together [75]. The average clustering coefficient of a graph with v vertices is signified as:

$$A_{VGC} = \frac{1}{N} \sum_{j=1}^N C_v, \quad (4)$$

where C_v denotes the local clustering coefficient with value

$$C_v = \frac{2E_v}{k_v(k_v - 1)}, \quad (5)$$

And E_v denotes the number of edges among the k_v neighbors of vertex v .

- d) *Average Path Length (Avg len)*: This graph measure is measured as the average of lengths of all the possible paths in the graph. It is experimentally calculated as [67]:

$$\text{Avg len} = \frac{1}{N(N-1)} \sum_{a \neq b} D(v_a, v_b), \quad (6)$$

where $D(v_a, v_b)$ represents the shortest length path between v_a and v_b vertices. And N signifies the total number of vertices in the graph.

- e) *Average Degree (AD)*: The Average Degree is defined as ratio of the total number of links (E) to the number of vertices in set V [75]. As each edge is present among two vertices and comes in the calculation of the degree of both the vertices, as a result AD of an undirected graph is represented as:

$$AD = 2 \frac{|E|}{|V|}, \quad (7)$$

- f) *Mean Strength*: It is a measurement of graph which is used to differentiate a monotonic change in the time series and denoted as [86]:

$$\overline{ms} = \frac{1}{N} \sum_{i=1}^N s_i, \quad (8)$$

where N signifies the total number of vertices and s_i epitomizes the strength of vertex v_i with Ew as an edge weight and is symbolized as:

$$s_i = \sum_{j=1}^N Ew_{ij}, \quad (9)$$

- g) *Variance of Degree Sequence*: This measure is used to evaluate the network heterogeneity, and is denoted as follows [90]:

$$\sigma = M^{-1} \sum_D d^2 M(D) - \langle d \rangle^2, \quad (10)$$

In the above equation, $\langle d \rangle^2$ is the average degree, and $M(D)$ are the number of vertices, which has degree d .

- h) *Average Weighted Degree (AWD)*: If $A_{N \times N} = \{a_{ab}\}$ symbolize an adjacency matrix with N number of vertices. Then $A_{ab} = 1$ if the edge exist from the vertex a to b otherwise 0. The weighted degree of vertex a is the sum of the weights of all the edges linked to vertex a and is symbolized by [94]:

$$wd_a = \sum_{j \in B(a)} w_{ab}, \quad (11)$$

where $B(a)$ indicates the neighborhood of vertex a and w_{ab} be a sign of the link weight among vertices a and b . The AWD is measured as the average of the total weights of the existing edges on all the vertices in the graph.

- i) *Loop coefficient*: Loop coefficient parameter is used to analyses the recurrence time distributions with the help of graph and represented as $H(I)$ [91]:

$$H(I) = \sqrt{k_{in}^i * k_{out}^i}, \quad (12)$$

where k_{in}^i and k_{out}^i are the number of inward and outward links from vertex i .

- j) *Graph Entropy*: This parameter is a complexity measure of a graph which is used to determine the information rate with the help of its structural content. For weighted graph, $E(v)$ represents the entropy of node v as follows [89]:

$$E(v) = - \sum_{h=1}^m a(v, h) \log(a(v, h)) \neq 0, \quad (13)$$

where

$$a(v, j) = \frac{w_{vj}}{\sum_{k=1}^m w_{vk}}, \quad (14)$$

In the above equation, w_{vj} denotes the weight of the edges among the v^{th} node and the j^{th} node and m signifies the number of nodes linked to the v^{th} node.

- k) *Degree Centrality (C_d)*: Degree Centrality measure is applied to quantify the total edges a node has. It is calculated as [74]:

$$C_d(v) = \frac{d(v)}{n-1}, \quad (15)$$

In the above equation no. 15, the $d(v)$ symbolizes the degree of the node v , n represents the total nodes in the graph and $n-1$ signifies the highest number of edges of one node.

- l) *Graph Density (GD)*: Graph Density is demarcated as the ratio of total present links to the total possible links between all the vertices [75]. It is calculated as:

$$GD = \frac{2E}{N(N-1)}, \quad (16)$$

where E denotes the number of links and N signifies the total nodes in the graph.

- m) *Transitivity (T)*: Transitivity is measured as the presence of a comparative number of triangle sets of network to the total number of linked triples of vertices. T is the fraction of triples [75]:

$$T = \frac{3 \times \text{number of triangles in network}}{\text{number of connected triples of vertices}}, \quad (17)$$

- n) *Global Efficiency (GE)*: Global Efficiency parameter of a graph is used to quantify about how efficient a graph is to interchange the information [75]. GE of a graph with V number of vertices is defined as:

$$GE = \frac{1}{N(N-1)} \sum_{a,b,a \neq b} \frac{1}{d_{ab}}, \quad (18)$$

in the above (18), the d_{ab} signifies the shortest path length between vertex a and vertex b .

- o) *Central Point Dominance (C_{pd})*: This parameter of a graph is used to measure the betweenness centrality of a graph or network [75]. The betweenness centrality B_v of a vertex v is defined as:

$$B_v = \sum_{ab} \frac{\sigma(a, v, b)}{\sigma(a, b)}, \quad (19)$$

where, $\sigma(a, v, b)$ denotes the number of shortest paths among nodes a and b that passes through node c , $\sigma(a, b)$ represents the total number of shortest paths among nodes a and b .

Central Point Dominance is measured as:

$$C_{pd} = \frac{1}{N-1} \sum_{a=1}^N (B_{\max} - B_a), \quad (20)$$

where $B_{\max}$ denotes the maximum betweenness in the graph and B_a is the vertex betweenness centrality [75].

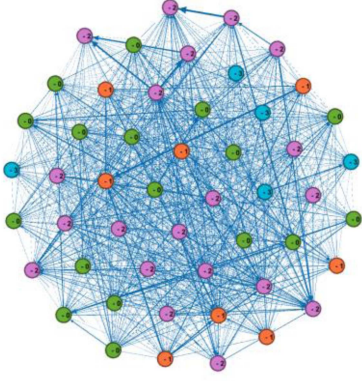


Fig. 1. Illustration of Graph of EEG dataset with colour of nodes represents the different groups.

- p) **Modularity (Q):** Modularity is measured as a complexity function for weighted graph to determine the quality of sub-division of the graph into the components or modules [88]. then the modularity is measured as:

$$Q = \frac{1}{2w} \sum_{ab} \left(M_{ab} - \frac{k_a k_b}{2w} \right) \delta(C_a C_b), \quad (21)$$

where Q symbolize the modularity, M_{ab} signify the weight of the links among vertex a and b . $k_b = \sum_b M_{ab}$ represent the sum of the weight of links associated with the vertex b . C_a denotes the community in which vertex a lies and w . $\delta(C_a C_b)$ is 1 if $C_a = C_b$ else 0.

When the module y combine into module z at that juncture the modularity gain is calculated as:

$$\Delta Q_{yz} = \left[\frac{\sum_{zn} + k_{y,zn}}{2w} - \left(\frac{\sum_{tot} + k_y}{2w} \right)^2 \right] - \left[\frac{\sum_{zn}}{2w} - \left(\frac{\sum_{tot}}{2w} \right)^2 - \left(\frac{k_y}{2w} \right)^2 \right], \quad (22)$$

where $\sum_{zn}$ symbolize the total weights of the links that come under module z ; $\sum_{tot}$ represents the total weights of the links that are incident to the node in the module; $k_{y,zn}$ denotes the sum of the weight of the links from the module y to module z ; k_y signify the sum of the weights of the links incident to vertex y ; w exemplifies the total weight of all the edges in the graph. Fig 1. demonstrate the community representation of the graph of small amount of EEG data set with. $E = \{1.049, -3.079, -6.122, -4.65, -4.453, -1.509, -2.883, -3.276, -2.294, 1.442, 3.993, 6.741, 8.606, 8.017, 5.662, 3.601, 4.778, 5.76, 2.619, 2.717, 3.601, 4.582, 7.428, 12.925, 15.476, 16.163, 13.612, 7.527, 3.012, 4.68, 5.564, 2.227, 3.797, 3.895, 2.717, -0.233, -6.51, -8.085, -6.907, -4.552, -3.079, -2, 1.049, 0.847, 0.847, -0.037, -1.313, -0.331, -0.233, -3.57\}$. The different colour of the nodes are illustrating the different groups or clusters in which the nodes belongs.

- q) **Assortativity (AC):** Assortativity of complex network is defined as correlation coefficient among the degrees of

all vertices on two opposite end of an edge or link and is measured as [75]:

$$AC = \frac{S_1 S_e - S_2^2}{S_1 S_3 - S_2^2}, \quad (23)$$

In the above (23), $S_e = \sum_{ab} A_{ab} k_a k_b$, $S_1 = \sum_a k_a$; $S_2 = \sum_a k_a^2$ and $S_3 = \sum_a k_a^3$. The more detailed information regarding assortativity is present in [75].

- r) **Motifs:** A motif is defined as a structural pattern described by the number of vertices or degree of nodes and the linkage among them and the number of points utilize among these vertices (called lag (λ)) [95]. To convert a time series $Z_{(i)}$ into a motif series $Z_m(i)$, the following equation is used:

$$Z_m(i) = \begin{cases} 1 & \text{if } Z_{(i)} < Z_{(i+\lambda)} < Z_{(i+2\lambda)} < Z_{(i+3\lambda)} \\ & < Z_{(i+2\lambda)}, \\ 2 & \text{if } Z_{(i)} < Z_{(i+\lambda)} < Z_{(i+2\lambda)} < Z_{(i+3\lambda)} \\ & < Z_{(i+2\lambda)}, \\ 3 & \text{if } Z_{(i)} > Z_{(i+\lambda)} < Z_{(i+2\lambda)} < Z_{(i+3\lambda)} \\ & < Z_{(i+2\lambda)}, \\ 4 & \text{if } Z_{(i)} < Z_{(i+\lambda)} < Z_{(i+2\lambda)} < Z_{(i+3\lambda)} \\ & > Z_{(i+2\lambda)}, \\ 5 & \text{if } Z_{(i)} > Z_{(i+\lambda)} < Z_{(i+2\lambda)} < Z_{(i+3\lambda)} \\ & > Z_{(i+2\lambda)}, \\ 6 & \text{if } Z_{(i)} < Z_{(i+\lambda)} < Z_{(i+2\lambda)} < Z_{(i+3\lambda)} \\ & < Z_{(i+2\lambda)}, \end{cases} \quad (24)$$

III. EPILEPSY DETECTION TECHNIQUES

In this section, we provide the overview about graph theory-based approaches and the epilepsy detection methods based upon the existing approaches. Most of the graph theory based epilepsy detection techniques are using the following sequence of steps:

- 1) Mapping of time series EEG signals into complex network
- 2) Topological features are extracted from the complex network of EEG signals.
- 3) Classification based upon the extracted feature sets.
- 4) Performance Measures.

Fig. 2 illustrates a schematic diagram of the complex network-based approaches for the identification of Epilepsy from EEG. For the mapping of time series EEG signals to complex network, each sample value of time series EEG signals are considered as the node of the complex network or graph. The links between different nodes are built on these specified criteria (as different researches used a different approach to find the link between the nodes). As the complex network is formed then the statistical features of complex network are extracted and features set are created. Afterward, with the help of classifiers and the extracted

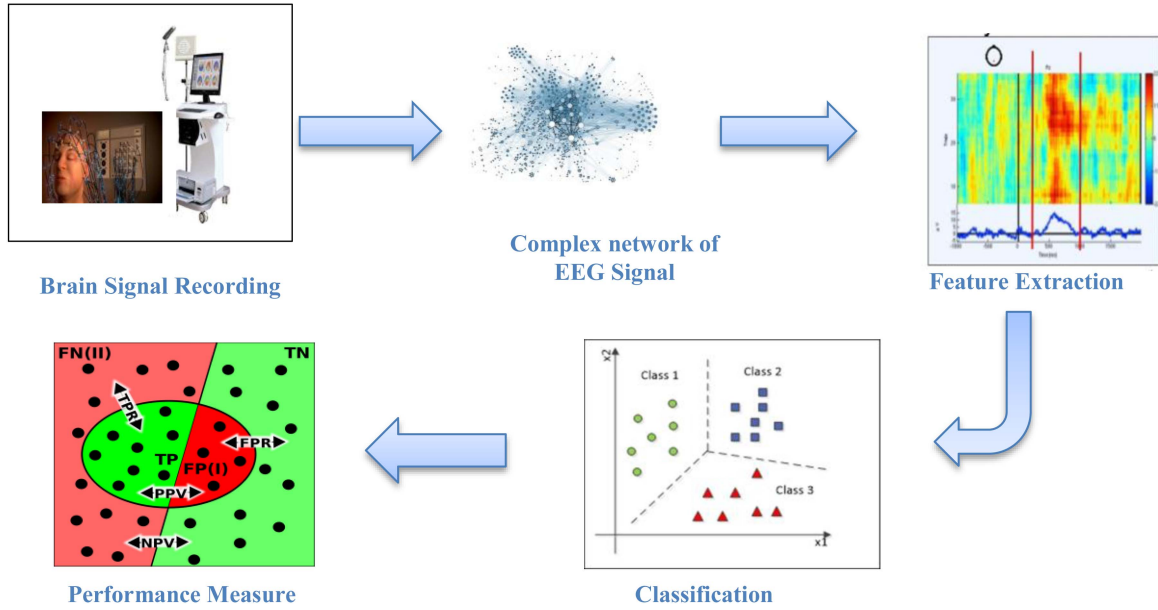


Fig. 2. Schematic representation of general sequence of steps followed by graph theory based approaches for epileptic seizure detection from brain EEG signals.

features set, the EEG signals are classified into their appropriate class (healthy or epileptic EEG signals). Based upon the requirement, some researchers used the feature selection method before classification. Once the classification is done then the performance of the anticipated techniques are assessed by the following measurements [53]:

$$\text{Sensitivity} = \frac{\text{True Positive}}{\text{True Positive (TP)} + \text{False Negative (FN)}}, \quad (25)$$

$$\text{Specificity} = \frac{\text{True Negative}}{\text{True Negative (TN)} + \text{False Positive (FP)}}, \quad (26)$$

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{FN} + \text{TN} + \text{FP}}, \quad (27)$$

There are some other performance metrics that are also used in the field of epilepsy detection i.e., Precision, Negative Predictive Value, False Positive Rate, False Discovery Rate, and Miss Rate, etc. [54].

Following are the brief introduction about graph theory-based approaches that are used in the epilepsy detection. In addition to this, the advantages and disadvantages of the various methods that had used these approaches are also discussed.

A. Visibility Graph Approach

In 2008, Lucas proposed the idea of the Visibility Graph Algorithm [55] to transform the time series into a network or graph on the basis of visibility character of geometry. The principle of VG is the theory of Euclidean plane, i.e., if each node denotes the point's position in the Euclidean plane, then the link between the allied nodes is only probable if there is visibility among them. This VG graph makes the visual analysis

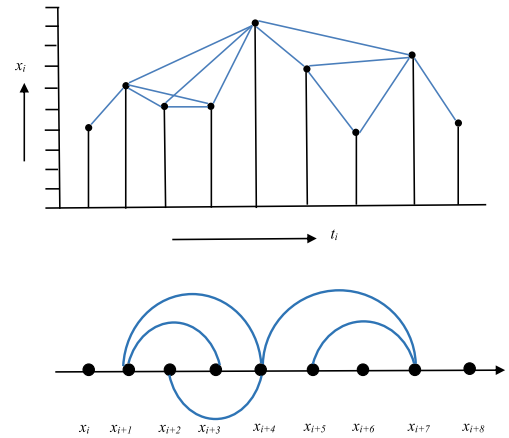


Fig. 3. Illustration of Visibility Graph [88].

of the structural patterns of the network, easy at various time scales of the microscopic level to the macroscopic level [56]. To understand the VG graph, let's assume $G(N, E)$ denotes a graph with N number of the node and E is the total edges. A time series is represented by $X = \{X_t\} (t = 1, 2, \dots, m)$ is a time-series. According to Lucas, if each data point (X_i) of the time series X is measured as a node (n_i) of the graph $G(N, E)$. The edges among the nodes of the graph $G(N, E)$ is only possible if they satisfied the following equation:

$$n_b < n_a + (n_c - n_a) \frac{t_b - t_a}{t_c - t_a}, c > b > a, \quad (28)$$

where, n_a, n_b and n_c are the nodes which relates with the data sample x_a, x_b , and x_c of the time series X and t_a, t_b and t_c are their corresponding time events. Fig. 3 exemplify the VG of small sample of time series data.

The VG technique plays a decisive role in several research fields like in the analysis of multifractal stochastic processes [57], econometrics field [58], [59], hurricane data [60], seismology [61], Human heartbeat dynamics [62], [63]. Ahmadlou *et al.* [64] first-time applied the VG in the brain signal analysis field to detect Alzheimer's syndrome. Later, based on VG, the following methods have been proposed for the analysis of EEG signals to detect the epilepsy disorder. The brief review about these VG based techniques are:

G. Zhu *et al.* [65] used VG algorithm to analyze the epileptic EEG signals. This research study used the public repository of Bonn University epileptic EEG database in which the EEG signal has a frequency of 173.61 Hz. Mean degree and Degree distribution features were used to distinguish between healthy, inter-ictal (period between epileptic seizures) and seizure state of EEG signals. The Degree distribution features were able to satisfy the power law whereas, the Mean degree parameter of epilepsy activity showed remarkable variance from normal EEG signals when used Linear Discriminant Analysis (LDA). On the other hand, the author used a small part of the whole database. In addition to this, the low degree nodes feature was not able to fully satisfy the power law.

X. Tang *et al.* [66] proposed the idea of using VG algorithm for epilepsy detection from high-frequency Electroencephalography (ECoG) signals with a frequency range of 100- to 200 Hz by using Butterworth Bandpass filter. The experimental data was collected from the Hospital of Peking University which comprises of ECoG signals of three epileptic patients. This research study focused on the comparative analysis between Graph index complexity (GIC), Sample Entropy and Lempel-Ziv Complexity parameters for the identification of Epileptic activity. The investigational results proved that the GIC feature was the most promising as compared to the other two features for the database used in the proposed study. Although, GIC was a good non-linear marker to detect epilepsy in case of high-frequency signals because it is sensitive to high-frequency fluctuations. However, GIC parameter also showed inconsistent results for some spatially independent channels.

C. Hao *et al.* [67] have done the analysis and prediction of the epilepsy seizure attack by using the VG algorithm. The study was based on two features: Average Path Length and Clustering Coefficient. According to the research study, Average Path Length does not perform any significant output for the classification of the epileptic signals whereas Clustering Coefficient feature produced quiet promising results. This research work was based on the small subset of Bonn university database. The drawback of this proposed methodology is that there were no performance measurements used for the evaluation of results. The research output was based on the visualization of the graph. As a result, it is hard to decide the classification performance of the proposed methodology without any performance evaluation measurement.

Ali *et al.* [68] develop a VG based algorithm to detect epilepsy seizure in which a particular size window was slide along the EEG signals to generate the segments. Then, each segment was considered as nodes and link between them was establish using VG algorithm. This research study claimed that the motifs of

the complex network could be considered as a marker for the analysis of epileptic seizure activity. During seizure activity, the nerve tissue discharge was spatially extended which in turn increases the rate of connectivity among vertex-pairs and the presence rate of pre-eminent motifs. The proposed method included with the shortcoming of the limitation of window size. If the window sized above 10, then the proposed methodology was not able to provide valuable information about EEG signals.

L. Wang *et al.* [69] performed EEG signal analysis using VG, HVG and DVG approaches to detect epileptic seizure in a patient with intellectual disability (ID) by collecting the data from Epilepsy Center Kempenhaeghe. The Mean degree (MD), Degree entropy (DE), Power-law degree power (DP), Assortativity coefficient (AC) and Average shortest path length (SL) parameters were extracted in the feature extraction part after mapping time series EEG data into three different graph(VG, HVG and DVG) to perform statistical analysis and 5 fold cross-validation using Support Vector Machine (SVM) classifier with a Gaussian kernel. The major finding of this research was the graph parameters of HVG provided more promising results to discriminate epileptic EEG from the non-epileptic EEG signals as compared to VG and DVG. It was author believed that the attained results might be not appropriate for automated epilepsy detection in actual clinical implementations. In addition to this, the seizure duration was very small in the EEG signals due to which the boundaries of the different stages of the seizure attack was difficult to distinguish.

M. Zeng *et al.* [100] proposed novel framework named entropy_of_visibility_heights_of_hierarchical_neighbors (EVHHN), for epilepsy detection. In EVHHN framework, with the help of visibility theorem, the visibility connections among three nearest neighbors are determine first. Then the visibility heights of the nearest-neighbors plays a crucial role for determine the four distinct categories of entropy associated in feature extraction. Finally, the SVM, LS-SVM, KNN and RF classifiers implemented on the Bonn Epilepsy database. The LS-SVM classifiers performed well in terms of accuracy measures as compared to others. The authors also claimed that the feature extracting time is very less.

Z. Gao *et al.* [96] developed an innovative technique for epilepsy detection based upon EEG signals analysis by combining the VG and adaptive optimal kernel time-frequency representation. Clustering coefficient, average degree and clustering coefficient entropy (CCE), were the parameters of the network that used for feature extraction. The classification results with higher accuracy performance reveals the importance of the complex network in the analysis of the Epilepsy detection from EEG.

B. Horizontal Visibility Graph Approach

Lacasa *et al.* [70] proposed a Horizontal Visibility Graph (HVG) Algorithm which is easier to understand at a geometrical level as well as analytical level as compared to VG algorithm. HVG is based upon the comparison of the amplitude of data sample points of EEG signals. According to HVG, the link between two points only exist if there is a direct horizontal

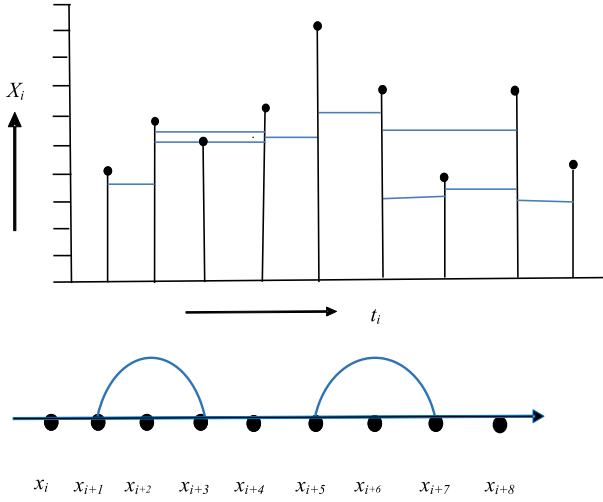


Fig. 4. Illustration of horizontal visibility graph.

line between the points without being intersected by any other point. In order to comprehend the HVG graph, let's assume $G(V, L)$ denotes a graph with V number of the vertex (node) and L number of links. A time series is represented by $X = \{X_t\} (t = 1, 2, \dots, n)$ is a time-series. According to Lacasa, if each data point (X_i) of the time series X is considered as a node (v_i) of the graph $G(V, E)$. The link among the nodes of the graph $G(N, E)$ is only established if they satisfied the following equation:

$$v_i, v_j > v_k, \forall k | i < k < j, \quad (29)$$

where, v_i, v_j and v_k are the nodes which relates to data points x_i, x_j and x_k of, the time series X and t_i, t_j and t_k are their corresponding time events. Fig. 4 illustrate the HVG between different time series data points.

The HVG technique has been used in the various field for the scientific purpose like in the analysis of heartbeat intervals [71], financial time-series [72] and river flow fluctuations [73], etc. The epileptic seizure detection based on HVG technique has been discussed below:

H. Liu *et al.* [74] proposed a method to distinguish epileptic seizure activity from non-seizure activity by using HVG algorithm and with the help of Degree Centrality parameter of a graph. The performance of the proposed methodology was check by the SVM classifier. It was claimed that Degree Centrality parameter of a HVG is a satisfactory marker for distinguishing ictal (during the seizure) from inter- ictal EEG signals. The major limitation of the proposed methodology was that the Degree Centrality parameter provided low performance as the size of the data sample of EEG signal increases. Moreover, the classification performance results were not as high as compared to state of the art presented in the research study [74].

P. Artameeyan *et al.* [75] developed a feature extraction based methodology using HVG for epileptic seizure detection. Three classifiers named KNN, MLPNN, and SVM were used to analyse the ten statistical characteristics (Average degrees, average-clustering coefficient, transitivity, assortativity, density, central point dominance, closeness, average short path, the global efficiency, and network diameters) of a complex network.

The authors achieved good classification accuracy results for the two databases used. However, the proposed methodology has some drawbacks as it is not appropriate for a large data sample of EEG signals. Furthermore, ten statistical characteristics of the complex network were used for classification which is time-consuming part as the number of feature sets increases, the feature extraction part takes more computation time. Even the p-value test for the two databases showed a big difference in value for some characteristics and also was not quite promising.

H. Liu *et al.* [76] have done seizure analysis with the help HVG algorithm and proposed the idea of the new feature named Improved Degree Centrality (IDC) to classify different EEG signals and achieved 96.5% classification accuracy. The major problem of the proposed method is that the newly developed feature IDC is not able to provide high-performance results when combined with the classifiers. In addition to this, a small part of the database was used, and on the basis of that, it is hard to depict how the proposed methodology will behave when used to differentiate different types of EEG signals.

Z. Gao *et al.* [97] proposed a novel technique based on the bridging of multiscale analysis as well as limited penetrable horizontal visibility graph. The two-dimensional feature vectors were generated by using average clustering coefficient and CCE. When these vector set employed to SVM classifier then 100% classification accuracy demonstrate that the proposed technique is effective for analyzing epileptic signal.

N. Ahmadi *et al.* [103] used different approaches to analyses the epileptic and psychogenic non-epileptic seizure EEG data. To analyses the functional network parameters, the Horizontal visibility graph approach also used and largest Eigenvalue, clustering coefficient, eigenvector, betweenness centrality, and strength are extracted measures of network used to evaluate the performance. The most promising results are obtained by using the beta-band of EEG frequency signals.

C. Time Series' Complex Network (TSCN) Approach

F. Wang [77] proposed an idea of mapping a time series into a complex network named it Time Series' Complex Network (TSCN). According to TSCN algorithm if $X = \{x_t\} (t = 1, 2, \dots, n)$ represents a time-series of length n . The time series is divided into various non-overlapping and equal sized length (k) cycles. Then each cycle is considered as a vertex of the complex network. Each cycle is representing as:

$$\left\{ C_a = (x_{(a-1) \times k + 1}, x_{(a-1) \times k + 2}, \dots, x_{(a-1) \times k + k}) \right\}, \quad |a = 1, 2, \dots, m \quad (30)$$

In some cases, the few end data sample points of time series are left so that $m \times k \leq n$. The distance between two cycles is measured by the subsequent equation which is based upon the Euclidean distance:

$$d_{ij} = \frac{1}{k} \times \sqrt{\sum_{h=1}^k (C_i(h) - C_j(h))^2}, \quad (31)$$

where, $C_i(h)$ is the h^{th} point of the cycle C_i . The edges between the vertices is based upon the following equation:

$$e_{ij} = \begin{cases} 1 & d_{ij} < \varepsilon \\ 0 & d_{ij} \geq \varepsilon \end{cases}, \quad (32)$$

where ε is a threshold value which is predetermined on the basis of some criteria. When $e_{ij} = 1$ then there is a edge (link) among vertex i and vertex j whereas, $e_{ij} = 0$ denote no link between them.

The following techniques have used the TSCN algorithm to detect the epileptic seizure:

F. Wang *et al.* [77] develop an algorithm to classify different EEG signals on the basis of feature extraction by using TSCN technique. The major finding of this proposed work is that the shape of Degree Distribution function can be used to classify the EEG signal because different dynamics of EEG signals have different shapes and it is hard to quantify that. This proposed study used Shannon entropy of Degree distribution feature to classify different EEG signals and achieved 96.25% classification accuracy. The major drawback of this proposed methodology is that there is no proper guidance has been provided to decide the value of the ε threshold. The value of ε is decided on the basis of executing the algorithm continuously until the promising accuracy does not achieved. Moreover, the selection of ε threshold value effect the network parameters which shows in turn effect the classification performance.

Latterly, F. Wang *et al.* [78] proposed a new idea of epilepsy detection by using Cluster Coefficient Distribution parameter of network and also have proposed some modification in the TSCN algorithm. In the new TSCN algorithm, the cycles of the TSCN was based upon the local maximum (or minimum) value of the EEG time-series. In addition to this, the distance between the two cycles was also based upon the minimum value. The ε threshold value is computed on the basis of changes of network density. The shortcoming of this proposed methodology is that the classification is only based upon the visual analysis of the clustering coefficient distribution of different EEG signals. There has been no use of proper classifier to check the classification performance, and visual analysis cannot be considered as an appropriate measurement for classification. Moreover, the reason behind of selecting a threshold value on the basis of particular elements of the network is also not clear. In addition to this, the classification accuracy of the proposed technique decreases with the decrease in the length of EEG signals.

F. Wang *et al.* [79] proposed a method to determine epileptic disorder from EEG signals by using traditional TSCN algorithm. The weighted mean value of degree distribution is calculated from degree distribution and considered as a feature for distinguishing ictal and inter-ictal EEG signals. The experiments were performed by considering different values of threshold ε and other parameters. The proposed method has fast computation speed and also achieved 96.50% classification accuracy which is better than their previous proposed methodologies. The main limitation of the proposed technique is that the threshold ε value selection criteria have not been provided. It is selected by executing the algorithm multiple times until the best accuracy does not achieve. There is no proper guidance has been provided

to decide the value of ε threshold. The classification outcomes are provided on the basis of the box plot diagram instead of any classifier. Moreover, the classification accuracy of the extracted feature named weighted mean value of degree distribution show inconsistency with ε as first it increases with the increase of threshold and later on it starts decreases with the increase of threshold.

D. Other Approaches Based on the Complex Network

K. A. Schindler *et al.* [80] analyzed the complex network of different EEG signals earlier, later and for the duration of epileptic seizure activity. Network path length and cluster coefficient were used as parameters for the research. The major findings of the proposed research work that the two parameters were shows transformation in the network topology by shifting from random to regular then again shifted to random. Before the starting of the seizure attack, the clustering coefficient showed an increase in value. The synchronization of neurons was decreased for the duration of seizure attack and shifted towards the increasing level before finishing of the seizure attack. The proposed study encompassed with the drawback of a threshold value for the selection of cross-correlation function and edge matrices.

C. Wike *et al.* [81] had applied graph measure to investigate the behavior of ictal activity in the epileptogenic networks. The out-degree parameter of the epileptogenic networks was used for the analysis. The major finding of the research study was that the out-degree parameter helped in the location of the seizure onset zone and gave information about the occurrence of the ictal movement in the brain. The research work had the limitation for the choice of optimal model order which varied according to the selection criteria.

S. Bialonski *et al.* [82] introduced the assortativity measure of the graph for the analysis of the epileptic EEG signals. Correlation coefficient and time lag were used to evaluate the interconnection of the signals. The major finding of this research study was that the positive degree-degree correlations feature of the network helps to characterize the seizure activity. In addition to this, the assortativity coefficient increased, and synchronization decreases for the duration of the epileptic seizure. The research study comprised with the shortcoming that the edge links are established using threshold value which was not certain.

Y. Ni *et al.* [83] have done an analysis of EEG signals in order to check how the small-world or scale-free topology of the brain network related to the epileptic seizure. The multiple-mass neural model was used to extract the neural network statistics for the duration of epileptic seizure. The main discovery of the proposed study was that the small world network and the scale-free network has a significant relation with the incidence of seizure attack.

G. Petkov *et al.* [84] used the computational model for understanding the underlying dynamics of the seizure activity in the functional network of EEG signals. The mean degree, degree variance, and local clustering coefficient were the measurements of the functional network used for the investigation. The research study concluded that the mean node degree parameter increased during epileptic seizure activity and degree variance parameter

also played an important role in distinguished different EEG signals. Whereas, the clustering coefficient feature had not performed really well with the proposed methodology.

W. Bomela *et al.* [102] introduce a new technique for epilepsy detection named as inferring the connectivity of networks (ICON). This technique helps to find the synchronization patterns by evaluating the coupling functions among multi-channel EEG data. The power and phase of the coupling functions helps to increase the robustness of the technique. The proposed technique provide high performance results as 93.6% of detection rate and 0.16 per hour of false positive rate.

E. Weighted Graph-Based Approaches

The above-cited techniques to detect epileptic seizure comprises of some shortcomings. But the common and major drawback that all of the above-mentioned techniques are having is the lack of edge strength in their proposed methodologies. According to R. Polikar, the preserving of weight information on the complex network helps to find more reliable result [85]. The strength of the edges plays an important role to analyze the existing information of the network. Different nodes of the network connect with each other through different intensities. The edge strengths help in to differentiate between strongly important and potentially weak node in the network which further assists to understand the underlying dynamical information about the network. The following techniques proposed their epileptic seizure detection methods by considering this important information about edge strength in their research study.

G. Zhu *et al.* [86] modified Horizontal Visibility Graph by introducing edge strength in it and named it as Fast Weighted Horizontal Visibility Graph (FWHVG). The mean degree and mean strength parameters of the FWHVG were extracted to check the classification performance with the help of KNN classifier. The major finding of the proposed methodology was that the network parameters extraction using FWHVG are faster and provides more satisfactory results as compared to Fast Fourier Transform and Sample Entropy technique. This research study achieved 100% accuracy for classifying the healthy and seizure activity EEG signals. In addition to this, the ictal and inter-ictal EEG signals classification results were also quite promising. The shortcoming of the proposed methodology is that the FWHVG does not include all of the data sample points of EEG signals. In addition to this, for the short length segment of EEG data sample points, the proposed technique provides less classification performance. Furthermore, the mean degree parameter of the FWHVG individually gives 82.5% average accuracy for classifying different types of EEG signals which is not quite promising.

Supriya *et al.* [87] introduce Weighted Visibility Graph (WVG) algorithm to detect epilepsy. A novel edge strength method was developed to find the weight between the different links of the network. The WVG is the improved version of the Traditional VG algorithm. This research work introduced a new feature named Average Weighted Degree (AWD) to detect epileptic syndrome by using a support vector machine (SVM) and Discriminant Analysis (DA) families of classifiers.

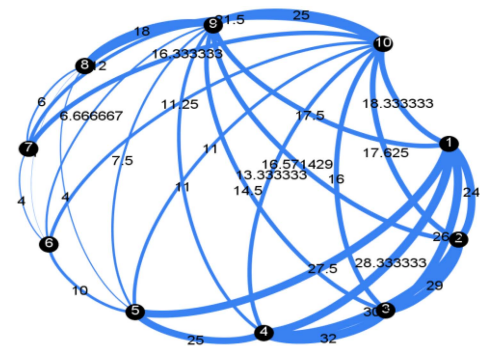


Fig. 5. Illustration of WVG of EEG signal with the edges represented the edge weight [94].

The major findings of the proposed technique were that the classification accuracy of the proposed technique did not affect by segmentation of EEG signals. The classification performance of the segmented and non-segmented EEG signals by using the AWD feature was very close to each other. The proposed methodology provided high-performance results for different test cases of EEG signals as compared to G. Zhu *et al.* [86]. Fig. 5. Illustrate the WVG of small amount of EEG signals during seizure activity with data points values = {100, 124, 153, 185, 210, 220, 216, 222, 240, 265}.

Supriya *et al.* [88] developed a new edge weight method to detect epilepsy from EEG signals by using WVG algorithm. This research study introduced the community structure based feature named Modularity for the detection of epilepsy syndrome and combined it with AWD feature for providing better results. The individual visual analysis of Modularity and AWD parameter proved that both of the features are independently competent to classify different types of EEG signals. To produce more efficient results, both the feature set was combined and achieved high-performance results by using KNN and SVM classifiers with different kernel functions. The major finding of the proposed technique was that the Modularity feature of the complex network is a good marker to discriminate between different types of EEG signals. This proposed technique achieved high classification performance as compared to Supriya *et al.* [77].

Z. Mohammadpoory *et al.* [89] proposed a new method to detect epilepsy by using WVG [87]. This research study introduced a new idea of extraction of entropy feature from WVG and named it as WVGE. The mean, standard deviation, maximum and minimum of WVGE were extracted as a features of the WVG. The performance of the proposed idea was checked by using four different classifiers: SVM, KNN, Decision tree and, Naïve Bayes. The Decision tree classifier provides promising results with 97% classification accuracy on the basis of three features (mean, standard deviation, and minimum) by differentiating three categories (named healthy, inter-ictal, and ictal) of EEG signals into their appropriate class.

M. Liu *et al.* [90] proposed a feature extraction method to detect epilepsy from EEG signals. The EEG signals were converted into Improved Transition Network (ITN) by introducing link strength into it. The variance of degree sequence of the network

TABLE I
EPILEPTIC SEIZURE DETECTION METHODS BASED UPON DIFFERENT APPROACHES OF GRAPH THEORIES

Technique Name	Authors	Database	Features	Classifier	Results or Classification accuracy
Visibility graph	G. Zhu et al. [65]	Bonn University epileptic database	Mean degree and Degree distribution	Linear Discriminant Analysis	Degree distribution features satisfied the power law
	X. Tang et al. [66]	Clinical database	Graph index complexity, Sample Entropy and Lempel-Ziv Complexity	N/A	GIC is sensitive to high-frequency fluctuations
	C. Hao et al. [67]	Bonn University epileptic database (Set C-E)	Average Path Length and Clustering Coefficient	N/A	Clustering Coefficient increased during the ictal period
	Ali et.al [68]	Bonn University epileptic database (Set A-E)	The degree of each vertex	N/A	During epileptic seizure, presence rate of pre-eminent motifs increased
	L. Wang et al. [69]	Epilepsy Center Kempenhaeghe	Mean degree, Degree entropy, Power-law degree power, Assortativity coefficient, and Average shortest path length	SVM with a Gaussian kernel	Degree entropy and power-law degree power features increased the performance when combined with other features.
	M Zeng et al. [100]	Bonn University epileptic database	four distinct types of entropies, i.e., Evh-1, Evh-2, Evh-3 and Sev-2-3	SVM, LS-SVM, KNN and RF	Less feature extraction time; LS-SVM provides high accuracy i.e. : 100% (A-E) 99.67% (AB-E) 98.50% (C-E) 98.35% (CD-E)
Horizontal Visibility Graph	P. Artameeyanant et al. [75]	1. Bonn University epileptic database 2. CHM-MIT database	Average degrees, average-clustering coefficient, transitivity, assortativity, density, central point dominance, closeness, average short path, the global efficiency, and network diameters	KNN, MLPNN, and SVM	Database 1: 100% (A-E) 98.80 % (ABCD)-E 98.60% (AB)-(CDE) 98.33% (A-D-E) 98.40% (AB)-(CD)-E 94.60% (A-B-C-D-E) Database 2: 100% (Non seizure-seizure)
	H. Liu et al. [74]	Bonn University epileptic database (Set D-E)	Degree Centrality	SVM	93.92% (sample length=1024) 92.98% (sample length=2048)
	H. Liu et al. [76]	Bonn University epileptic database (Set D-E)	Improved Degree Centrality (IDC)	N/A	96.50%
	N. Ahmadi et al. [103]	Ghent University Hospital in Belgium	Largest Eigenvalue, clustering coefficient, eigenvector, betweenness centrality, and strength	k-NN, DT, Neural Network, RF, NB, SVM and Gradient Boosting	beta-band of EEG frequency provides efficient results.
TSCN	F. Wang et al. [77]	Clinical database	The entropy of the degree distribution	N/A	96.25%
	F. Wang et al. [78]	Bonn University epileptic database (Set D-E)	The partial sum of cluster coefficient distribution	N/A	94.50%
	F. Wang et al. [79]	Bonn University epileptic database (Set D-E)	weighted mean value of degree distribution	N/A	96.50%
Other approaches based on Graph-Theory	K. A. Schindler et al. [80]	Clinical database	Network Path Length and Cluster Coefficient	N/A	During a seizure, the parametric topology shifted to regular and synchronization decreases.
	C. Wike et al. [81]	Clinical database of intractable neocortical-onset epilepsy	Out-degree	N/A	The out-degree is a good biomarker for seizure onset zona.
	S. Bialonski et al. [82]	Clinical database	Assortativity Coefficient	N/A	Assortativity coefficient increased during epileptic seizure.
	Y.Ni et al. [83]	Clinical database	Regular, Random, Small World, and Scale-Free Networks.	N/A	The small world network and scale-free network are closely related to epileptic seizure activity.
	G. Petkov et al. [84]	Clinical database	Mean Degree, Degree Variance, and Local Clustering Coefficient	N/A	Mean node degree significantly increased for the duration of the epileptic seizure.
Weighted Graph-based	G. Zhu et al. [86]	Bonn University epileptic database	Mean Degree and Mean Strength	KNN	100% (A-E) 93.0% (D-E) 95.4% (A, B, C, D), E

TABLE I
CONTINUED

	Supriya et al. [87]	Bonn University epileptic database	Average Degree	Weighted	SVM, LDA, and QDA	100% (A-E) 95% (B-E) 94.5% (C-E) 98.5% (D-E)
	Supriya et al. [88]	Bonn University epileptic database	Modularity Average Degree	and Weighted	SVM and KNN	100% (A-E) 97.25% (B-E) 98.25% (C-E) 93.25% (D-E)
	Z. Mohammad poory et al. [89]	Bonn University epileptic database (Set A-D-E)	mean, standard deviation, maximum and minimum of entropy		KNN and SVM, NV, and DT	97.0% (A-D-E)
	M. Liu et al. [90]	Bonn University epileptic database (Set D-E)	variance of degree sequence		N/A	98.5% (D-E)
	M. Liu et al. [91]	Bonn University epileptic database (Set D-E)	loop coefficient		N/A	98.5% (D-E)
	Supriya et al. [92]	Bonn University epileptic database	Average Degree and Average Degree	Weighted and Average	Naive LDA, SVM, Bayes, QDA,	100% (A-E) 97.0% (B-E) 98.50% (C-E) 95.50% (D-E)
	S. Modak et al. [104]	Bern Barcelona EEG database.	Average Coefficient, Average Density, Average Local Efficiency and Global Efficiency	Clustering	SVM	92.5%
	H Zhang et al. [93]	Bonn University epileptic database (Set D-E)	average degree weight		linear classifier	99% (D-E)

was used as the feature extracted from ITN to classify EEG signals. The experimental results proved that the epileptic EEG signals based network has a low variance of degree sequence and better consistency as compared to EEG signals of a healthy person. The accuracy of the proposed method was 98.5%. Trials were performed on the different data sample length of EEG signals, and the classification results were much closer to each other for different length.

M. Liu *et al.* [91] presented an innovative feature extraction method to classify different EEG signals. The EEG signals were firstly transformed into transition network. On the basis of a number of transitions from one segments to another, the edge weight was calculated, and finally, a weighted and directed graph was constructed from different EEG signals. Loop coefficient was used as an extracted parameter from weighted transition network to classify ictal and inter-ictal EEG signals. The major finding of the proposed method was that the loop coefficient of the Epileptic EEG signals based network was small because the chaos of epileptic EEG signals was weak and recurrence time was also small. The proposed method achieved 98.5% classification accuracy.

Supriya *et al.* [92] introduce a method to differentiate Epileptic EEG signals from the healthy person EEG by introducing weight on the horizontal visibility graph and named it as Weighted Horizontal Visibility Graph (WHVG). The proposed technique was tested on all of the five classes of different EEG signals of Bonn University Epileptic EEG database. The Average Degree and Average Weighted Degree were the two feature which extracted from WHVG to check the classification performance using different machine learning classifiers: Naive Bayes, Linear Discriminant Analysis, and Quadratic Discriminant Analysis, Support Vector Machine (SVM) classifier with Linear, Rbf and Polynomial kernel function. The classification

results for the four test cases were closed to each other, but SVM classifier with polynomial kernel provides high performance as compared to others. The classification accuracy for Epileptic EEG signals from the healthy person EEG was 100%.

H. Zhang *et al.* [93] developed a novel technique to classify different EEG signals using a fusion of features of HVG and WHVG. In this research study, the average degree square feature was extracted from HVG and average weighted degree were extracted from WHVG. Then the feature fusion technique of network was proposed by addition of these two features and named it as average degree weight. With the help of linear classifier, the newly developed feature, i.e., average degree weight achieved 99% classification accuracy to differentiate ictal EEG signals from inter-ictal EEG signals.

S. Modak *et al.* [104] applied weighted visibility graph for epilepsy detection. Average Clustering Coefficient, Density, Average Local Efficiency and Global Efficiency measures of network extracted. SVM classifier applied and 92.5% classification accuracy has been achieved for focal and non-focal epileptic EEG data.

IV. DISCUSSION AND PERSPECTIVES

The automated epilepsy detection based upon graph theory is definitely a vast and promising approach. It is also one of the active topic in seizure detection. The graph theory based seizure detector can be helpful in preventing the impairment during epilepsy diagnosis and definitely advance patient treatment [88], [89]. The main focus during the development of seizure detector is to explore the hidden dynamics of different EEG signals under diverse circumstances. The clinically diagnosis of epilepsy involves the complete history in addition to neurological checkups and blood tests. In some cases, cerebrospinal fluid is also tested to check the biochemical disproportions. Therefore, automatic

detection of epilepsy based upon EEG signals is on great demand. The early diagnosis of epilepsy not only ameliorate the quality of the life of a subject with epilepsy but also provide safety from physical dangers.

There are still some challenges in the development of automated epilepsy detection devices like dependability in the real-time emanation of data, portable batteries, and vast amount of data, maintenance price, and research funding.

Table I depicts a comparative report of various graph-based methods in epilepsy detection. It is evident from this research study that graph theory plays a crucial role in the analysis of epileptic seizure from EEG signals. The topological features of a network is a good biomarker for seizure detection. This current study also investigates that the weighted graph approach is one of the most promising approach for seizure detection as it helps to distinguish between important and less informative links in the EEG graph. The theoretical analysis of the research studies also evidences that the clustering coefficient, mean degree and average shortest path length features does not show consistency in their results. Their performance depends upon the type of the graph theory approach used, i.e., they behave in a different way when used with a different approach of graph theory. In addition to this, the literature study also explores that their computational complexity is effectively low when used in real-time applications [99].

On the basis of the above discussions and details, we are expecting that this research work covers most of the graph theory based techniques which are widely used in the epileptic seizure detection. We believed that this current work will be a helpful resource for the researchers to get information about graph theory based seizure detection techniques.

V. CONCLUSION

The main objective of this research study is to provide knowledge to the researchers and clinicians about the state-of-the-art in the field of epilepsy detection based upon graph theory. This research work will assist them in the research directions of complex network-based seizure detection approaches. This paper includes the brief information about that various approaches, in which, the time series EEG signals are first transformed into the complex network and then graph based measurements are used to analyze them. By introducing the concept of adding weight in the complex network, play a significant role for EEG signal analysis as well as epilepsy detection as it provides information about the strong and weak link in the graph of EEG signals. Although, there are various graph-based techniques are available to detect epilepsy from EEG graph, but still, there is essential to develop some more efficient methods which can determine the gap present in the EEG patterns. The EEG data analysis is a challenging issue because of its complexity, diversity, sheer volume (in size and dimension) and the rich in context. The automated epilepsy detection techniques can be helpful in the diagnosis of epilepsy, neurosurgical treatment, in the deterrence of accidents and reduce risk. In addition, it will assist the neurologist in the analysis for EEG data and also helpful in

the establishment of patient care and improve the quality of life of the epileptic patient.

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